INCH-POUND

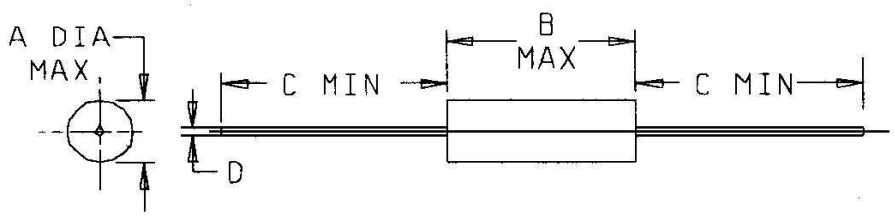
MIL-PRF-27/286C
w/AMENDMENT 2
20 June 2024
SUPERSEDING
MIL-PRF-27/286C
w/AMENDMENT 1
17 March 2017

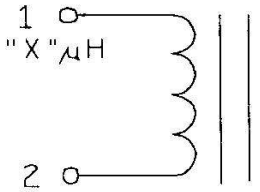
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE) INDUCTORS, POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.





WV: 50 VOLT PK
ALTITUDE: 70,000 FT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Marking shall be on the side of the case.
5. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.

AMSC N/A FSC 5950

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Dim	Inches	mm
A	.375 MAX	9.53 MAX
B	.500 MAX	12.7 MAX
C	1.25 MIN	31.8 MIN
D	.032 Dia	0.81 Dia

FIGURE 1. Dimensions and configurations – continued.

REQUIREMENTS:

Electrical ratings: See table I.

TABLE I. Electrical ratings.

Dash no. 1/	Inductance µh (1-2) +30 percent -5 percent 2/ 3/	DC current amps (1-2) max 3/ 4/	DC resistance ohms (max) (1-2) 3/	Voltage (1-2) 3/
01	120	1.6	0.14	1 volt 20 kHz
02	300	1.0	0.35	1 volt 20 kHz
03	500	0.75	0.56	1 volt 20 kHz
04	1200	0.50	1.40	1 volt 20 kHz
05	3000	0.30	3.50	1 volt 20 kHz
06	5000	0.25	5.60	1 volt 20 kHz
07	30	3	.035	1 volt 20 kHz
08	50	2.5	.056	1 volt 20 kHz

1/ Qualification test and approval to M27/286-06, shall be sufficient to grant qualification approval to M27/286-01 through M27/286-08.
2/ The inductance is measured with 0 A dc applied to (1-2) and at the specified voltage across (1-2).
3/ When numbers are in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.
4/ At rated dc current, inductance will exceed 80 percent of rated inductance.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Epoxy.

Terminals: .032 inch oxygen free copper leads.

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Weight: 4.5 grams, maximum.

Altitude: 75,000 feet, maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202-211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 100 volts rms.

At reduce barometric pressure: 100 volts rms.

Vibration (high frequency): MIL-STD-202-204.

Temperature rise: +45°C with 1 volt rms, 20 kHz, current as specified in table I applied to (1-2) at an ambient temperature of +60°C maximum.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/286-(dash number from table I).

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2024-021)

Review activities:
Army - AR, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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